The 1-3 vs. 2-2 Egg-Drop Strategy: Which Constant Wins?

1. The problem

You have **4 identical eggs** and a building with **N floors**. There is a critical floor F^* such that an egg dropped from a floor $\leq F^*$ survives, and from a floor $> F^*$ it breaks. You want to determine F^* in the **worst case** with the **fewest drops**.

Two natural strategies are on the table:

- **1-3 approach:** Use one egg to narrow the search to an interval, then solve a 3-egg problem inside that interval.
- **2-2 approach:** Partition the floors into superfloors of size F, solve a 2-egg problem on the superfloors, then a 2-egg problem on the F floors of the critical superfloor.

Both require a number of drops that grows like $N^{1/4}$. The question is: **which one has the smaller constant?**

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2. The exact asymptotic constant for the optimal 4-egg strategy (the 1-3 approach)

The 1-3 approach is, in fact, the **optimal** 4-egg strategy when the intervals are chosen optimally. The key combinatorial fact is:

> With d drops and k eggs, the maximum number of floors you can test is
>
> $$M(d,k) = \sum_{i=0}^{k} \binom{d}{i}.$$

For $k = 4$,

$$M(d,4) = 1 + d + \frac{d(d-1)}{2} + \frac{d(d-1)(d-2)}{6} + \frac{d(d-1)(d-2)(d-3)}{24}.$$

For large d, the dominant term is

$$M(d,4) \sim \frac{d^4}{24}.$$

To be able to test N floors we need

$$\frac{d^4}{24} \geq N \quad \Longrightarrow \quad d \geq (24N)^{1/4}.$$

Thus the **optimal 1-3 strategy** uses

$$\boxed{d_{1-3} \sim 24^{1/4}\, N^{1/4}}.$$

Numerically,

$$24^{1/4} = (2^3 \cdot 3)^{1/4} = 2^{3/4}\, 3^{1/4} \approx 1.6818 \cdot 1.3161 \approx 2.213.$$

So the constant for the 1-3 approach is

$$\boxed{C_{1-3} = 24^{1/4} \approx 2.213}.$$

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3. The constant for the 2-2 approach

The 2-egg problem is classical: for M floors the optimal worst-case number of drops is the smallest d such that

$$\frac{d(d+1)}{2} \geq M,$$

so asymptotically

$$d_2(M) \sim \sqrt{2M}.$$

Now apply this twice:

1. **Superfloor phase:** Treat the N floors as $\frac{N}{F}$ superfloors, each of size F.
   Number of drops:
   $$d_1 \sim \sqrt{2\,\frac{N}{F}}.$$
2. **Inside the critical superfloor:** You have F floors.
   Number of drops:
   $$d_2 \sim \sqrt{2F}.$$

Total drops:

$$D(F) \sim \sqrt{\frac{2N}{F}} + \sqrt{2F}.$$

Minimise over $F > 0$. Let $x = \sqrt{F}$. Then

$$D(x) = \frac{\sqrt{2N}}{x} + \sqrt{2}\, x.$$

Setting the derivative to zero:

$$-\frac{\sqrt{2N}}{x^2} + \sqrt{2} = 0 \quad \Longrightarrow \quad x^2 = \sqrt{N} \quad \Longrightarrow \quad F = \sqrt{N}.$$

At this optimum,

$$D_{\min} = \frac{\sqrt{2N}}{N^{1/4}} + \sqrt{2}\,N^{1/4} = \sqrt{2}\,N^{1/4} + \sqrt{2}\,N^{1/4} = 2\sqrt{2}\,N^{1/4}.$$

Thus the **2-2 strategy** has

$$\boxed{C_{2-2} = 2\sqrt{2} \approx 2.828}.$$

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4. Comparison

Strategy	Constant in front of $N^{1/4}$	Approx. value
1-3 (optimal 4-egg)	$24^{1/4}$	**2.213**
2-2 (two 2-egg phases)	$2\sqrt{2}$	**2.828**

The 1-3 approach is better: its constant is about

$$\frac{2.828}{2.213} \approx 1.278,$$

i.e. the 2-2 approach needs roughly **28 % more drops** in the asymptotic limit.

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5. A note on the paper's bound

The paper *Egg Drop Problems: They Are All They Are Cracked Up To Be!* proves the **upper bound**

$$P_1(k) \le \lceil k\,N^{1/k} \rceil.$$

For $k = 4$ this gives $4N^{1/4}$. That bound is **not tight**; the exact asymptotic constant for the optimal 4-egg strategy is $24^{1/4} \approx 2.213$, not 4. The 2-2 constant $2\sqrt{2} \approx 2.828$ lies between these two numbers, so the paper's bound alone would not tell you which strategy is better—you need the exact constant to see that the 1-3 approach wins.

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